

3 Methods and Materials

3.1 Methodological Approach

Following the clear, phased presentation style used in recent surveys on LLMs and mental health, for example Ghorbian and Ghobaei Arani [9], this section explains TrustMind as a structured methodological pipeline rather than as an informal product narrative. The primary objective is to measure, under matched controls, to what extent RAG improves the trustworthiness, reliability, and explainability of LLM wellbeing assessments compared with a standalone LLM, while shipping a non diagnostic student check in built over allow listed public guidance.

The work proceeds in five phases, shown in Figure 1: problem and research question design; dual pipeline system architecture; allow listed knowledge base construction and indexing; seed reproducible synthetic evaluation data; and fair dual arm evaluation with an explicit trust layer. Three artefacts result from this process: a live product, a curated BM25 (and optionally FAISS) knowledge base, and $n = 500$ reliability metrics.

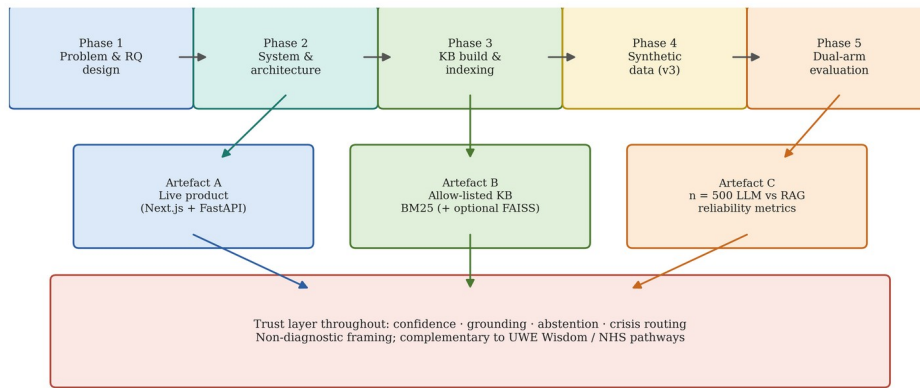


Fig. 1. TrustMind methodological approach: five phases from research question design to dual arm evaluation, producing the live product, curated knowledge base, and reliability artefacts.

3.1.1 Defining Research Inquiries

Table 2 states the inquiries that organise the method, mirroring the survey style TQ tables used in [9], adapted here to an empirical dual arm product study.

Table 2. Research inquiries guiding the TrustMind method.

Index	Question	How this section addresses it
TQ1	What system design compares LLM only against LLM plus RAG fairly?	Dual pipeline, same labels and model family, controlled retrieval mode (Sections 3.2, 3.3, 3.6)

TQ2	How is ethically evaluable data formed without Reddit scrapes?	Synthetic Wellbeing v3.0 generator (Section 3.4, Fig. 3)
TQ3	Why BM25, FAISS, and RRF, and when is each used?	Literature linked justification and decision flow (Section 3.5, Fig. 4)
TQ4	How are trustworthiness and explainability operationalised beyond accuracy?	Calibration, grounding, abstention, source cards, crisis rules (Sections 3.2, 3.5)

3.2 Problem Formulation and System Architecture

TrustMind maps a confirmed user check in, plain text that may optionally be derived from speech, an image, or a PDF after user confirmation, to one of five non diagnostic theme indicators: depression, SuicideWatch, Anxiety, bipolar, and offmychest. These labels are theme proxies compatible with the SWMH schema [1]; they are not diagnoses.

The frontend is built with Next.js and deployed on Vercel; the backend is built with FastAPI and deployed on Render, with PostgreSQL available for optional saved check ins. The browser never holds provider keys. The endpoint POST /api/v1/analyse accepts a pipeline_mode of llm, rag, or auto. Live generation uses multi provider routing (Groq, then Gemini, then OpenAI); the offline reliability table fixes the model to gpt-4.1. Multimodal files are converted to text, confirmed by the user, and never written into BM25 or FAISS. Mode A (llm_pipeline.py) uses a classification prompt only. Mode B (rag_pipeline_service.py) always retrieves BM25 passages and may optionally fuse FAISS results via Reciprocal Rank Fusion [2] (Section 3.5). A trust layer then applies calibration, grounding, abstention, and independent crisis routing before the interface shows reflections and, on the RAG arm, grounded sources. Figure 2 summarises this pipeline end to end.

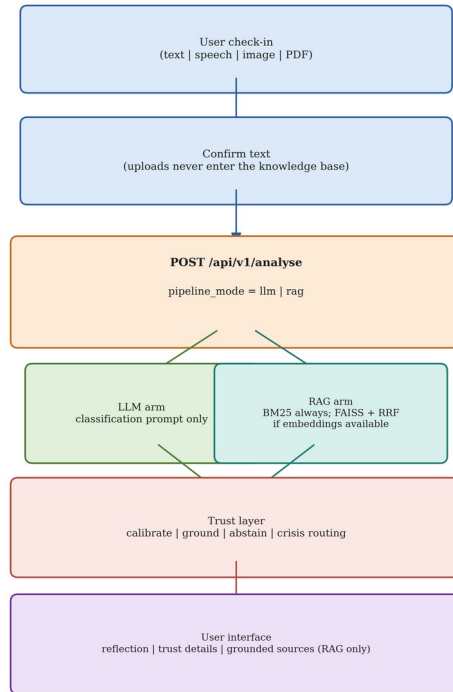


Fig. 2. TrustMind analyse pipeline: confirmed check in text enters either the LLM only or the LLM plus RAG path, then the trust layer, then the interface. Uploaded files never enter the knowledge base indexes.

3.3 Knowledge Base Construction

Phase 3 builds an allow listed knowledge base only (approved_sources.csv, over 110 rows): NHS style guidance, charity and student resources, UWE wellbeing pages, crisis charity pages, and selected peer reviewed PDFs; there is no open web crawl. Sources are cleaned to Markdown and chunked into 500 word windows with a 100 word overlap. The indexes comprise BM25 (primary) and an optional FAISS index. The production checkout contains on the order of 90 cleaned documents and 836 chunks. This implements the gap identified in Section 2 around public guidance rather than DSM or ICD style corpora [3]. Table 3 summarises the materials pipeline.

Table 3. Knowledge base construction stages.

Stage	Artefact / tool	Design choice
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Allow list	approved_sources.csv	Manual curation; no crawl
Acquire and clean	Collector and HTML/PDF cleaner	Rate limited fetch or local PDF
Chunk and index	500/100 word windows; BM25; optional FAISS	BM25 primary in production

3.4 Synthetic Dataset Formation

Phase 4 answers TQ2 from Table 2. Following supervisor advice against using Reddit sourced SWMH data, for which individual research consent could not be established, evaluation instead uses TrustMind Synthetic Wellbeing v3.0 (N = 2500; splits of 1600, 400, and 500 for train, validation, and test; seed 42). Figure 3 shows how the texts are formed entirely with software rather than being scraped or copied from SWMH posts.

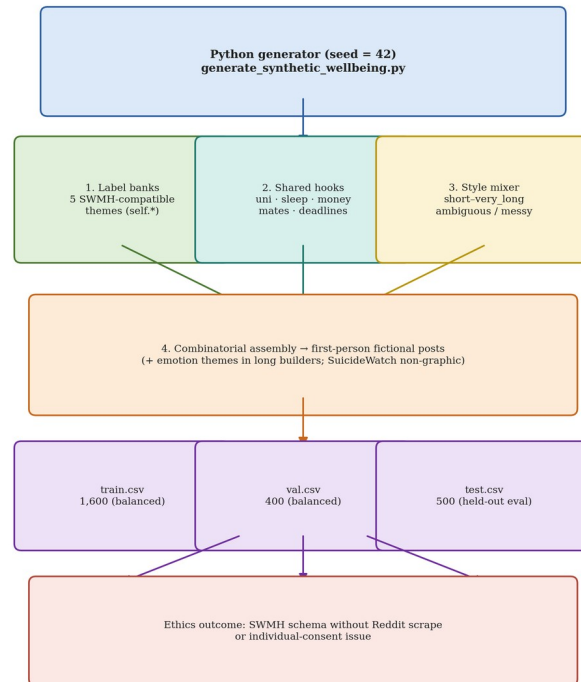


Fig. 3. Synthetic Wellbeing v3.0 formation: a seed reproducible Python generator combines label banks, shared student life hooks, and a length and style mixer into balanced train, validation, and test CSV files.

Step by step formation technology (research/generate_synthetic_wellbeing.py).

1. Label banks: five SWMH compatible theme fragment libraries (depression, SuicideWatch, Anxiety, bipolar, offmychest).

2. Shared hooks: everyday student cues (university, sleep, money, mates, deadlines) that appear across classes so a model cannot rely on a single keyword.
3. Style mixer: roughly 22 percent ambiguous, 18 percent messy, 12 percent short, 18 percent medium, 20 percent long, and 10 percent very long entries, covering sparse posts through narratives of around 150 to 800 words.
4. Combinatorial assembly: templates combined with random composition under seed 42; the longer builders add themes such as loneliness, heartbreak, stress, mixed feelings, hope, anger, guilt, and numbness, plus non diagnostic energy swings; SuicideWatch entries stay non graphic and use help seeking language.
5. Export: balanced CSV files; evaluation preprocessing scrubs URLs and @handles and caps entries at 512 words. The live product is not trained on this set; it exists purely for offline dual arm inference.

This formation approach directly fills the ethics gap highlighted in surveys and dual arm mental health RAG work that rely heavily on social media corpora [1,3,9].

3.5 Retrieval Techniques: BM25, FAISS, and RRF

Phase 3 and Phase 5 retrieval design together answer TQ3 from Table 2. Section 2 showed that RAG can improve provenance in principle [4], yet it can also add noise or hurt accuracy in some settings [3,6,7], while healthcare RAG reviews warn about poorly matched retrieval more generally [8]. TrustMind therefore decomposes RAG into three distinct technologies, each governed by an explicit when to use rule, shown in Figure 4.

3.5.1 Why These Techniques Were Chosen

BM25, always on within the RAG arm, fills the gap left by standalone LLMs, which are fluent but unauditible, and by dense only stacks that fail without embedding APIs. Lexical overlap matches student phrasing to NHS and charity headings, stays inside the allow list unlike free web search, and avoids the diagnostic DSM or ICD knowledge bases used in Ge et al. [3].

FAISS with embeddings, kept optional, fills the sparse only gap: paraphrases such as drained and empty versus low mood need semantic neighbours to be found [8]. It uses text-embedding-3-small together with a FAISS IndexFlatIP index when available.

RRF, optional and used alongside FAISS [2], fills the fusion gap: BM25 and dense scores sit on incomparable scales. RRF, with a constant k of 60, merges ranks without a trained re ranker and reduces the risk of one channel dominating the fused list, a noise pattern noted specifically when vanilla RAG hurts performance [3,6,7].

Table 4 links each of these choices back to the specific gap it closes.

Table 4. Technique selection linked to literature gaps.

Component	Chosen	Gap in prior or alternative technique	How TrustMind fills it
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Grounding	Allow listed RAG	Free web search; DSM/ICD style RAG [3]	Provenance [4] over public NHS/Mind style guidance
Sparse path	BM25 always on RAG arm	Dense only fails without embeddings; LLM only has no sources	Grounded sources without embeddings
Dense path	FAISS with embeddings (optional)	Sparse only misses paraphrases [8]	Semantic recall when the API is healthy
Fusion	RRF ($k = 60$) when both lists exist	Incompatible BM25/dense scores; naive merge noise [2,8]	Rank based fusion, top-k truncation
Eval data	Synthetic Wellbeing v3.0	Reddit SWMH consent issues [1,3,9]	Seed 42 template synthesis; balanced $N = 2500$
Trust	Calibrate, abstain, ground	Overconfident raw LLM scores [5]	Explicit trust user experience layered on retrieval

3.5.2 When FAISS and RRF Are Used, and How They Help

BM25 runs on every LLM plus RAG request, including the $n = 500$ reliability arm. FAISS and RRF are optional enrichments layered on top, and Figure 4 sets out exactly when each path is taken.

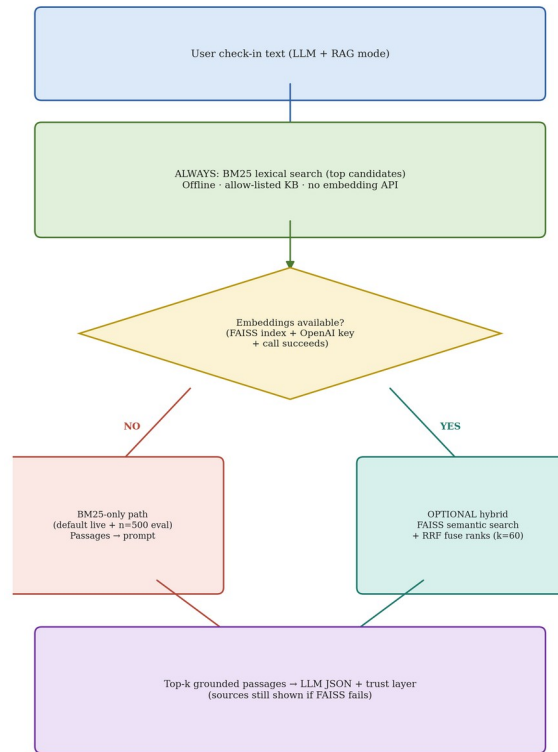


Fig. 4. Retrieval decision flow: BM25 always runs on the RAG arm; FAISS and RRF run only when embeddings succeed, otherwise the system falls back to BM25 only so that grounded sources remain available.

BM25 only is the default live path today: if embeddings are missing, rate limited, or fail outright, the system still returns allow listed passages, and `retrieval_mode` is often reported as `bm25_only`. In the hybrid path, once a FAISS index is present on disk and an OpenAI embedding call succeeds, FAISS returns its top 20 neighbours, BM25 contributes its own top 20, RRF fuses the two ranked lists, and the result is truncated to the product's top k of 5. FAISS improves paraphrase recall; RRF improves balanced fusion between the two channels. The soft fallback design means grounding is never dropped solely because dense retrieval failed.

3.6 Experimental Protocol

Phase 5 answers TQ1 from Table 2 for the reliability dimension specifically. Figure 5 and Table 5 state the fair dual arm controls used for the primary reliability metrics; the full protocol write up is in `research/results/rq_answer_summary.md`.

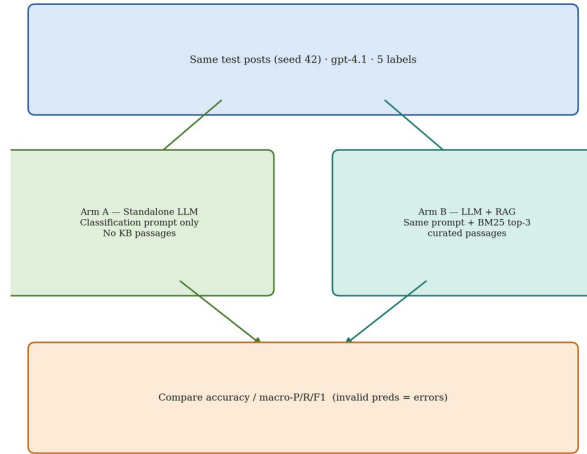


Fig. 5. Fair dual arm protocol: identical held out posts and model; Arm A has no retrieved passages; Arm B adds BM25 top 3 curated passages; the two arms are compared on accuracy and macro precision, recall, and F1.

Table 5. Fair dual arm experimental controls.

Control	Value
Test set	Synthetic wellbeing test.csv, n = 500, seed 42
Model	gpt-4.1 via the local TrustMind API proxy
LLM arm	Same five class prompt; no passages
RAG arm	Same prompt plus BM25 top 3 curated passages
Metrics	Accuracy; macro precision, recall, F1 (invalid predictions counted as errors)

The product's live retrieval configuration (BM25 always on, with optional FAISS and RRF, returning a top 5) is not identical to the reliability configuration reported here (BM25 only, top 3); both configurations are stated honestly wherever results are presented.

3.7 Delivery

Short, supervisor aligned iterations delivered the dual pipeline and trust interface first, then the synthetic dataset and offline ablation, and finally deployment hardening. The assessable artefacts match Figure 1: the live system, the committed knowledge base indexes, and reproducible evaluation scripts.

References

1. Ji, S., Li, X., Huang, Z., Cambria, E.: Suicidal Ideation and Mental Disorder Detection with Attentive Relation Networks. *Neural Computing and Applications* (2021). <https://doi.org/10.1007/s00521-021-06208-y>
2. Cormack, G.V., Clarke, C.L.A., Buettcher, S.: Reciprocal Rank Fusion Outperforms Condorcet and Individual Rank Learning Methods. In: *SIGIR 2009*, pp. 758–759 (2009)
3. Ge, Z., Hu, N., Wang, Y., et al.: Survey and Experiments on Mental Disorder Detection via Social Media: From Large Language Models and RAG to Agents. *arXiv:2504.02800* (2025)
4. Lewis, P., et al.: Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. In: *NeurIPS* (2020)
5. Guo, C., Pleiss, G., Sun, Y., Weinberger, K.Q.: On Calibration of Modern Neural Networks. In: *ICML*, pp. 1321–1330 (2017)
6. Xu, S., Yan, Z., Dai, C., Wu, F.: MEGA-RAG: a Retrieval-Augmented Generation Framework with Multi-Evidence Guided Answer Refinement for Mitigating Hallucinations of LLMs in Public Health. *Frontiers in Public Health* 13, 1635381 (2025)
7. Zhang, X., Liu, H., Zhang, Q., Ahmed, B., Epps, J.: SpeechT-RAG: Reliable Depression Detection in LLMs with Retrieval-Augmented Generation Using Speech Timing Information. In: *Findings of ACL 2025*, pp. 10019–10030 (2025)
8. Neha, F., Bhati, D., Shukla, D.K.: Retrieval-Augmented Generation (RAG) in Healthcare: a Comprehensive Review. *AI* 6(9), 226 (2025)
9. Ghorbian, M., Ghobaei-Arani, M.: Large Language Models for Mental Health Diagnosis and Treatment: a Survey. *Artificial Intelligence Review* 59, 9 (2026). <https://doi.org/10.1007/s10462-025-11418-0>